PROJECT SUMMARY

Smart Load Management and Backup Protection

Real-Time Voltage/Current Monitoring · Relay-Based Load Switching · Automatic Overload Protection

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1. Abstract

This project presents the design and implementation of a low-cost, microcontroller-based battery monitoring and protection system built around the Arduino Uno platform. The system continuously measures the voltage and current delivered by a 12 V, 7.5 Ah sealed lead-acid (SLA) battery using an INA219 digital current/voltage sensor and displays the readings in real time on a 16×2 LCD. A four-channel relay module allows connected DC loads to be switched under program control, while a ten-segment LED bar-graph gives an at-a-glance visual indication of system status, and a piezo buzzer provides an audible alarm. When the measured current exceeds a pre-set safety threshold, the controller flags an overcurrent condition, displays a "WARNING! OVERCURRENT" message on the LCD, sounds the buzzer, and can de-energize the relay to isolate the load and protect the battery. The prototype was assembled on breadboards for testing and was validated by loading the battery until the current exceeded the safe limit, confirming that the protection routine triggers correctly and reliably. The project demonstrates a practical, low-cost, and expandable approach to battery health monitoring and overcurrent protection suitable for small UPS, solar, and backup-power applications.

2. Project Background

Rechargeable batteries — particularly sealed lead-acid (SLA) batteries used in UPS units, solar backup systems, and small DC power banks — are vulnerable to damage from overcurrent, over-discharge, and unmonitored operation. An unexpected current surge, a short-circuited load, or a fault in a connected

appliance can draw more current than the battery or wiring is rated for, leading to overheating, reduced battery life, or even fire hazards. In most low-cost backup power systems, however, batteries are connected directly to the load with little or no active monitoring, leaving faults undetected until damage has already occurred.

Microcontroller platforms such as the Arduino Uno, combined with inexpensive digital sensor modules like the INA219, make it feasible to add real-time monitoring and automatic protection to such systems at very low cost. By continuously sampling voltage and current and comparing the readings against safe operating limits, a simple embedded controller can warn the user of abnormal conditions and automatically disconnect the load using a relay before damage occurs. This project was undertaken to design and build a working prototype of such a system, demonstrating how basic embedded electronics can be used to make small-scale battery-powered systems safer and more reliable.

3. Objectives

- **To design a circuit that measures the real-time voltage and current of a battery using the INA219 sensor module.**
- **To display live voltage, current, and system status information on a 16×2 LCD.**
- **To implement relay-based switching so that DC loads can be connected to or disconnected from the battery under program control.**
- **To provide visual (LED bar-graph) and audible (buzzer) indicators of system/load status.**
- **To automatically detect an overcurrent condition and respond with a warning message, alarm, and load disconnection to protect the battery.**
- **To provide manual rocker switches and a push button for user override and mode selection.**
- **To assemble and test the complete system on a breadboard prototype and validate its behaviour under both normal and fault (overcurrent) conditions.**

4. Required Components

The system was built using readily available, low-cost modules commonly used in Arduino-based embedded projects. Table 1 summarises the major components used and the role each one plays in the system.

#	Component	Qty	Function in the Project
1	Arduino Uno (ATmega328P)	1	Main controller — reads sensor data, runs the monitoring/protection logic, and drives all outputs
2	INA219 Current/Voltage Sensor Module	1	I2C sensor that measures the battery's bus voltage and load current in real time
3	4-Channel 5V Relay Module (SRD-05VDC-SL-C)	1	Switches DC loads / battery lines on and off under Arduino control, and isolates the load during a fault
4	16×2 Character LCD (blue backlight)	1	Displays operating mode, live voltage/current readings, and warning messages
5	LED Bar-Graph (10 LEDs + resistors)	10	Visual indicator of load/battery status level
6	Piezo Buzzer+ Red LED	1+1	Audible alarm with red light signal during an overcurrent/fault condition
7	Rocker Switches (SPST)	3	Manual ON/OFF control of individual output channels
8	Opto Coupler(PC817)	1	Detect presence of main grid to switch to main power
9	12 V, 7.5 Ah Sealed Lead-Acid (SLA) Battery	1	The battery/power source being monitored and protected
	12V Adapter (Variable)		Represent main source, As the project designed for DC for safe operation
10	Breadboards (full-size + mini) & Jumper Wires	several	Prototyping platform for all connections
11	Resistors (assorted)	assorted	Current-limiting for LEDs and signal conditioning

Table 1: List of major components used in the project

5. Methodology

5.1 System Design

The Arduino Uno forms the core of the system. It communicates with the INA219 sensor module over the I2C bus to obtain live bus-voltage and current readings from the SLA battery. These readings are processed in software and written to the 16×2 LCD, which shows the current operating mode together with the live voltage (V) and current (C) values. Digital output pins from the Arduino drive the four-channel relay module, which switches the battery/load lines, while additional outputs drive the LED bar-graph and the piezo buzzer for status and alarm indication. Three rocker switches and a push button are wired to digital input pins to allow the user to manually enable/disable channels and select the operating mode.

5.2 Circuit Assembly

The prototype was assembled on a set of breadboards mounted on a foam board for stability. The Arduino Uno, INA219 sensor, and relay module were placed on the main breadboard cluster, with the 16×2 LCD mounted separately and connected via a ribbon of jumper wires for the data and control lines. The 10-LED bar-graph, each LED fitted with its own current-limiting resistor, was built on a separate breadboard to provide a clear visual scale. The three rocker switches were mounted along the front edge of the board for easy manual access, and the 12 V/7.5 Ah SLA battery was connected as the power source under test, with its output routed through the relay module and the INA219 sensor so that the current drawn by the load could be measured directly.

5.3 Software Logic

On start-up, the Arduino initialises the I2C bus, the LCD, and the input/output pins. In the main loop, the controller periodically reads the bus voltage and current from the INA219 sensor and updates the LCD with the selected mode label together with the live V and C readings. The measured current is continuously compared against a pre-set safe threshold. Under normal operation the LCD displays the mode and readings as usual (e.g. "M:bat-A 0.70A" / "V:11.8 C:0.638A"). If the current exceeds the threshold, the controller switches the display to a warning screen ("WARNING! OVERCR" / "V:11.8 C:0.630A"), activates

the buzzer, updates the LED bar-graph, and drives the relevant relay channel to disconnect the load, protecting the battery from sustained overcurrent. The rocker switches and push button are polled in the same loop to allow the user to change channels or reset the system.

6. Results and Discussion

The assembled prototype is shown in Figure 1. All modules — the Arduino Uno, INA219 sensor, relay module, LCD, LED bar-graph, buzzer, and manual switches — were successfully interconnected and powered from the 12 V/7.5 Ah SLA battery under test.

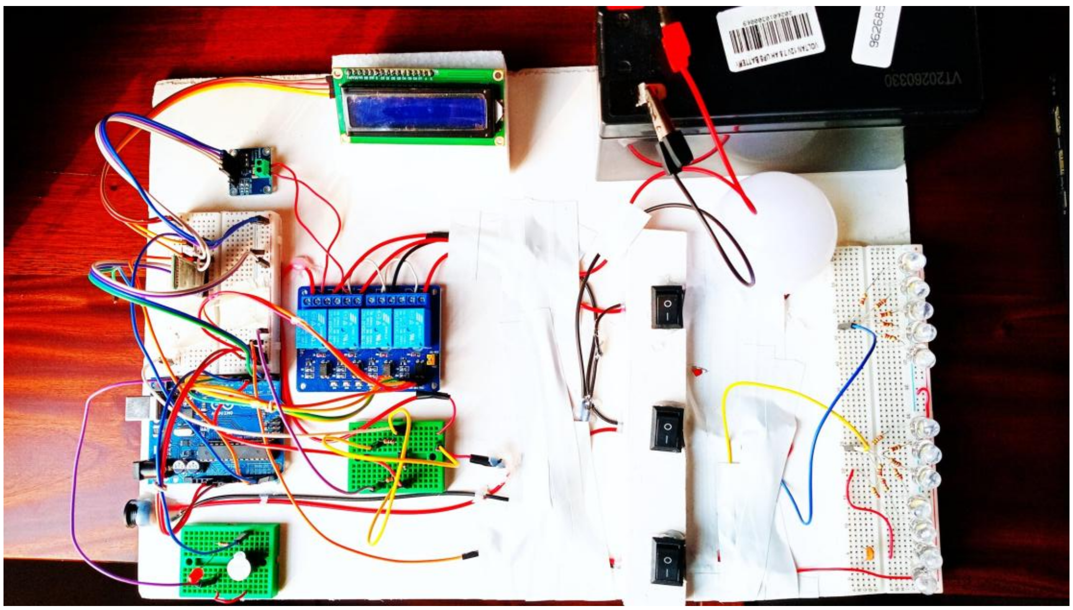

Figure 1: Complete hardware prototype of the battery monitoring and overcurrent protection system

During normal operation, the LCD correctly displayed the live battery voltage and current — for example, a reading of V:11.8 V and C:0.638 A was recorded under a steady load, close to the 0.70 A reference shown for the selected mode ("M:bat-A 0.70A"). This confirms that the INA219 sensor and the Arduino's data-acquisition routine were able to measure and refresh the readings reliably in real time.

To validate the protection function, the load current was increased until it crossed the safe operating threshold. As shown in Figure 2, the system correctly detected the fault condition: the LCD switched to the alert screen reading "WARNING! OVERCR" together with the live readings (V:11.8 V, C:0.630 A), confirming that the fault-detection and warning routine executed as intended. Together with the buzzer and LED bar-graph, this gives the user immediate visual and audible feedback that the system has entered a protective state, and allows the relay to isolate the load before the battery is subjected to sustained overcurrent.

Figure 2: LCD showing the overcurrent warning triggered during testing

Overall, the results show that the prototype achieves its main objectives: it monitors battery voltage and current continuously, presents the data clearly on the LCD, and responds automatically to an overcurrent fault with a clear warning and (via the relay) load isolation. Being built on breadboards, the current prototype is best suited to bench testing; the exposed wiring and lack of a rigid enclosure would need to be addressed before the system could be deployed in a real UPS or solar backup unit. The manual rocker switches also indicate that, alongside the automatic protection logic, the design retains a layer of manual control, which is useful for testing individual channels.

7. Conclusion and Future Work

The project successfully demonstrates a working Arduino-based battery monitoring and overcurrent protection system, combining real-time sensing, clear on-board display, and automatic fault response in a single low-cost prototype. Testing confirmed that the system accurately reports live voltage and current and reliably triggers a warning and protective action when the current exceeds the safe limit.

Future improvements could include migrating the circuit from breadboards to a soldered PCB and enclosure for durability, adding data logging or wireless (Wi-Fi/Bluetooth) reporting for remote monitoring, incorporating over-voltage and deep-discharge protection alongside overcurrent protection, and calibrating the current threshold against the specific battery's datasheet ratings for more precise protection.

WHY IT IS GOOD

Feature	Benefit
Priority load shedding	Critical loads always protected
Bluetooth app control	Remote monitoring and control
Auto + Manual modes	Flexible operation
Real time sensing	Accurate INA219 measurements
Relevant to Bangladesh	Daily power outage solution